

Web Development Bootcamp: Week 1 Lesson Notes

Week 1: HTML Foundations

Day 1: HTML Basics & Structure

The internet consists of web clients requesting static files from servers. Your desktop browser reads these nested text arrays and translates them directly into human-readable visual layouts. Let us use the analogy of ordering custom meals inside a fast-food restaurant hub:

The Client (Browser): The customer sitting at a table placing an explicit request for a hamburger.

The Server (Remote Computer): The central kitchen storage processing raw asset data ingredients.

The Internet Protocol (HTTP): The fast waiter fetching requests and carrying fresh trays back and forth.

- **Tags:** Structural layout indicators notifying the engine how to parse textual contents (e.g., **<h1>** creates a top-level heading block, **<p>** defines paragraph body lines).
- **<!DOCTYPE html>**: An essential browser system alert note flags down: 'Heads up computer, clean modern HTML5 incoming!'
- **<html>**: The master structural root inventory shipping storage crate enclosing all internal code nodes. It is the root document scope wrapping.
- **<head>**: The hidden operations brain module holding language properties, text descriptors, and titles invisible to end-users.
- **<body>**: The real physical main display arena room canvas where every visible visual item is rendered live.
- **Semantic Tags (<header>, <main>, <section>, <footer>)**: Instead of storing shoes, textbooks, and loose video game equipment inside one single unorganized bag, we use explicit compartments so engines like Google search indexing can crawl and sort layouts accurately

- **HTML Attributes:** The custom operational adjectives or configuration properties detailing precisely how that element node functions. They are nested entirely inside the opening tag brackets (e.g., '**src**' links graphic assets, '**href**' targets hyperlink navigation pathways).
- '**src**' (Source): The explicit address link pointer instructing the system layout exactly where to fetch the media file from data warehouses.
- '**alt**' (Alternative Text): Essential accessibility notes describing image contexts out-loud for screen-readers or during server downtime events.
- **Self-Closing Tags:** Since media tag links like images don't wrap physical text characters directly inside text frames, they snap shut internally with zero trailing tag markers required!

Anatomy of an HTML Element:

```
<tagname attribute="value">Content goes here</tagname>
```